Ovarian Cancer

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Microbiome Signatures, Microbiome Medicine

Abstract. Ovarian cancer is often diagnosed at late stages, making early detection crucial. New research highlights the potential of microbiome therapies.

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Overview

Ovarian cancer (OC) remains one of the most deadly cancers affecting women, with an estimated 320,000 new cases diagnosed worldwide annually, making it the eighth most commonly diagnosed cancer. It also accounts for over 200,000 deaths each year, reflecting its high lethality. The disease is often diagnosed at advanced stages (stage III and IV) due to the lack of early symptoms, leading to a poor prognosis.[1][2] Around 70% of ovarian cancer cases are diagnosed in these late stages, significantly reducing survival rates. The five-year survival rate for ovarian cancer remains dismally low, with an average of 30-45% in most countries.[3] This poor prognosis is largely attributed to the late detection of the disease and the frequent development of chemoresistance, particularly to platinum-based chemotherapy, the standard treatment for ovarian cancer.[4] Ovarian cancer is characterized by high heterogeneity, with various histological subtypes and molecular profiles. The most common type, epithelial ovarian cancer (EOC), makes up approximately 90% of all ovarian cancer cases.[5] Emerging research has shown the critical role of the microbiome in ovarian cancer progression, with recent studies linking both the vaginal and gut microbiota to carcinogenesis. Dysbiosis, or microbial imbalance, in these areas may contribute to inflammation, immune evasion, and tumor progression.[6] The gut microbiome, in particular, is known to influence the immune system and metabolic pathways, potentially playing a significant role in the development of ovarian cancer.[7] Repurposed non-oncology drugs are showing promise in ovarian cancer treatment.[8] These drugs have demonstrated antiproliferative, pro-apoptotic, and antimetastatic properties, offering a more accessible and cost-effective alternative to traditional chemotherapy.

Associated Conditions

Ovarian cancer is closely associated with several gynecological and systemic conditions that increase the risk of developing the disease. One of the most notable associations is with endometriosis, a condition where tissue similar to the uterine lining grows outside the uterus, often on the ovaries.[9] Women with endometriosis face a significantly increased risk of developing ovarian cancer.[10] Studies show that the relative risk of ovarian cancer in women with endometriosis can be as high as 4.2 times compared to the general population.[11] Specifically, women with ovarian endometriosis have a higher likelihood of developing endometrioid and clear cell subtypes of ovarian cancer. Other significant risk factors for ovarian

cancer include genetic mutations, particularly in the BRCA1 and BRCA2 genes.[12] Women carrying mutations in these genes are at a substantially higher risk, with up to 15% of ovarian cancers being attributed to inherited genetic mutations.[13] These mutations lead to an increased risk of both ovarian and breast cancers, with the lifetime risk of developing ovarian cancer for BRCA1 carriers reaching 40-60%, and for BRCA2 carriers, the risk is about 10-30%.[14] Obesity and hormonal therapies, including the use of hormone replacement therapy (HRT), are additional risk factors for ovarian cancer. Studies suggest that obesity increases the risk of ovarian cancer due to chronic inflammation and the impact of adiposity on hormonal pathways. Women using HRT, especially after menopause, may experience an elevated risk of developing ovarian cancer due to prolonged exposure to estrogen.

Causes

The causes of ovarian cancer are complex and multifactorial, involving genetic, hormonal, and environmental factors. A major hypothesis for ovarian cancer initiation is hormonal exposure, particularly to estrogen. Extended exposure to estrogen, through factors like early menarche, late menopause, and the use of oral contraceptives or HRT, can increase the risk of ovarian cancer.[15][16] Ovulation frequency has also been implicated in ovarian cancer development, as frequent ovulation increases the number of times the ovarian surface epithelium is damaged, leading to an accumulation of genetic mutations.[17] Genetic mutations, particularly in BRCA1 and BRCA2, are well-established causes of ovarian cancer.[18] These genes are involved in DNA repair mechanisms, and mutations in them lead to genomic instability, increasing the risk of cancer development. Lynch syndrome, caused by mutations in MLH1, MSH2, and other mismatch repair genes, is another hereditary condition that predisposes individuals to ovarian cancer.

Emerging theories suggest that microbiome dysbiosis may also play a pivotal role in ovarian cancer pathogenesis. The vaginal and gut microbiota have been shown to influence immune responses and metabolic pathways that can either promote or inhibit cancer development. Dysbiosis in the vaginal microbiota, characterized by a depletion of Lactobacillus and an increase in anaerobic bacteria, has been associated with higher rates of ovarian cancer.[19][20] Several case–control studies have identified that prior sexually transmitted infection, particularly *Chlamydia trachomatis*, increases ovarian cancer risk.[21] It is hypothesized that pelvic infections may contribute to ovarian carcinogenesis by inducing DNA damage, oxidative stress, and pro-inflammatory responses.[22] This link is still under exploration, but the growing body of evidence suggests that microbiota-related factors may contribute significantly to the disease's onset and progression.

Causal Theory	Evidence
Hormonal Exposure (Estrogen)	Increased estrogen exposure is linked to higher ovarian cancer risk. Studies have shown that prolonged exposure to estrogen, particularly in the form of menopausal hormone therapy, can increase the risk of developing ovarian cancer. [23]
Genetic Mutations (BRCA1/2)	Mutations in BRCA1/2 are linked to a significant increase in ovarian cancer risk.[24]
Microbiome Dysbiosis	Vaginal and gut microbial imbalances contribute to ovarian cancer risk.[25]

Diagnosis

Diagnosing ovarian cancer remains one of the greatest challenges in oncology due to its often silent progression and the lack of effective screening tools. Currently, serum biomarkers like CA125, HE4, and CEA are used in diagnosis, but these markers lack the sensitivity and specificity required for early-stage detection.[26][27] CA125, for instance, is elevated in only about 50-60% of early-stage cases.[28][29] Imaging techniques like ultrasound and CT scans are often used to detect masses in the ovaries, but they cannot reliably distinguish between benign and malignant growths. In recent years, liquid biopsy has shown promise as a non-invasive diagnostic method.[30] Additionally, the role of the microbiome in ovarian cancer diagnosis is being increasingly recognized. Studies have demonstrated that microbiome signatures, particularly from vaginal and gut microbiota, could serve as potential diagnostic biomarkers, offering a new avenue for early detection.[31] This approach may lead to the development of microbiome-based diagnostic tests, allowing for less invasive, earlier detection of ovarian cancer.

Primer

Ovarian cancer is one of the deadliest gynecological cancers, primarily due to its late-stage diagnosis and asymptomatic nature. The disease often presents with vague symptoms, leading to diagnoses at advanced stages with a low survival rate. High-grade serous ovarian cancer (HGSOC), the most common and aggressive subtype, shows significant metabolic and genetic dysregulation. Recent research highlights the crucial role of the microbiome, particularly vaginal and gut microbiota, in cancer progression and treatment responses. Metabolomics, the study of metabolites in biological samples, has identified biomarkers related to OC, including altered glycolysis and lipid synthesis, which are hallmarks of cancer cell survival and proliferation. Additionally, metallomics, which examines the role of metals like zinc, copper, cadmium, lead, and chromium, has underscored their involvement in cancer progression.[32] Dysregulation of essential metals, along with toxic metals like cadmium and lead, can cause DNA damage, oxidative stress, and hormonal disruption, contributing to tumor growth and chemoresistance. Understanding the intersections between the microbiome, metallomics, and metabolic alterations is key to improving early detection and treatment strategies for ovarian cancer.

Metallomic Signatures

Heavy metals like zinc, copper, lead, cadmium, and chromium are essential for various biological functions, but their dysregulation can promote tumorigenesis. These metals participate in key cellular processes such as oxidative stress, DNA repair, and apoptosis. Elevated levels of toxic metals, such as cadmium and lead, have been linked to ovarian cancer progression through genotoxicity, oxidative stress, and disruption of estrogen signaling.

▶ **What are the Metallomic Signatures of Ovarian Cancer**

Metabolomic Signatures

Metabolomics has emerged as a promising approach for understanding the biochemical pathways involved in ovarian cancer. Metabolomic studies have identified specific metabolic changes that occur in ovarian cancer cells, which support rapid tumor growth and resistance to treatment.[46] These metabolic reprogramming features include alterations in glycolysis, lipid metabolism, and amino acid metabolism, which are critical for cancer cell survival.[47][48] For instance, the Warburg effect, which involves the shift

from oxidative phosphorylation to glycolysis for ATP production, is frequently observed in ovarian cancer cells. This metabolic shift allows cancer cells to meet the increased energy demands required for rapid cell division.[49][50] Additionally, changes in lipid metabolism, such as increased synthesis of phospholipids, have been observed in ovarian cancer tissues. These alterations contribute to the structural integrity of cancer cell membranes and support cell proliferation.[51] Certain metabolites, such as N1,N12-diacetylspermine, polyamines, and betaine, have been consistently elevated in both plasma and ovarian cancer tissues, making them potential biomarkers for diagnosis and prognosis.[52][53] The identification of these metabolites through metabolomic profiling holds great promise for improving early detection and monitoring the progression of ovarian cancer.

Interventions

Primary, first-line ovarian-cancer care still centers on maximal cytoreductive surgery (up-front or interval) coupled to platinum–taxane chemotherapy; despite this, recurrence after standard platinum regimens is common, which is why maintenance and targeted add-ons have become mainstream.[54][55] Bevacizumab (anti-VEGF) is an approved targeted agent in the frontline/maintenance setting.[56] Improved progression-free survival in large phase-III programs, as well as the pairing of olaparib with bevacizumab (PAOLA-1), boosted outcomes, especially in homologous recombination-deficient disease. Building on the backbone, drug repurposing is an active strategy: metformin exerts indirect anticancer effects by lowering insulin/IGF signaling and direct effects via complex-I inhibition with downstream AMPK activation, mTOR/PI3K restraint, and dampening ERK/STAT3, mechanisms that align with chemosensitization and anti-angiogenesis in preclinical ovarian models.[57] In vivo and in vitro work showed metformin suppressed adhesion, invasion, migration, and reduced neovascularization and macrophage infiltration in ovarian-cancer models, consistent with antimetastatic activity, while a 2024 population cohort found post-diagnosis metformin associated with improved survival in diabetics (aHR 0.71 for ovarian-cancer–specific mortality), encouraging, though not yet definitive for all comers.[58][59] Nutritional care is another pillar that directly affects tolerance and outcomes: up to 70% of women with ovarian cancer present malnourished and 40% have low muscle mass at diagnosis; early, individualized nutrition counseling (including in prehabilitation), oral nutrition support, and immunonutrition are feasible, reduce symptom burden, and may improve survival, an area urgently needing larger prospective trials but already important in practice.[60]

Microbiome-targeted interventions (MBTIs) for ovarian cancer

Multiple groups have confirmed that ovarian tumors, like several other cancers, harbor an intratumoral microbiome detectable by 16S/ITS profiling, with microbial signatures that may have diagnostic/prognostic value, while requiring rigorous contamination control.[61] In parallel, bacteria themselves are being developed as therapeutics: intratumoral or tumor-seeking microbes (or engineered strains) can deliver cytotoxics or immunomodulators directly in situ (e.g., anaerobes like *Clostridium novyi-NT* and engineered *E. coli/Bifidobacterium* platforms), illustrating a concrete path for "intratumoral microbiota therapy."[62] Mechanistically, the ovarian-cancer–microbiome crosstalk appears to run through inflammation and metabolism: gut dysbiosis can raise systemic IL-6 and Other cytokines, activating JAK/STAT3 and NF-κB; in murine models, TLR5-responsive microbiota elevated IL-6 and accelerated ovarian tumor progression,

whereas depleting symbionts abrogated the effect, linking microbial patterns to tumor-promoting inflammation.[63] On the protective side, commensals in the female tract and gut, *Lactobacillus* biofilms and lactic acid in the vagina, *Clostridium* spp., and *Akkermansia muciniphila* reinforce barriers and tune antitumor immunity.[64] Microbially derived short-chain fatty acids (SCFAs) can suppress inflammatory chemokines (e.g., CCL20, IL-8) within the tumor milieu.[65] Translationally, this opens four levers to correct dysbiosis and amplify antitumor immunity: (1) probiotics to restore barrier organisms and recalibrate cytokine tone. (2) microbiota supplementation via fecal microbiota transplant (FMT) or strain-level enrichment to ignite $CD8^+$ T-cell activity and reverse tumor-promoting consortia.[66] (3) deliberate use of intratumoral/engineered bacteria as "living drugs." (4) boosting SCFAs (by diet/prebiotics or direct postbiotics) to quiet NF-κB/STAT3-linked inflammation and enhance effector T-cell function. In ovarian-cancer models, adding *Akkermansia* through FMT increased $CD8^+$ T-cell activity and flipped tumor behavior toward antitumor effects, supporting a rational, microbiome-targeted add-on to standard therapy.[67]

Intervention	Mechanism	MBTI Status
Probiotics (e.g., *Lactobacillus*, *Bifidobacterium*, potentially *A. muciniphila*)	Rebuilds epithelial and mucosal barriers; *Lactobacillus* biofilms block pathogen adhesion and its lactic acid increases IL-1RA and suppresses TNF-α, reducing local inflammatory tone.[68] In the gut, commensals educate immunity—*Clostridium* spp. expand Tregs; *A. muciniphila* expands T-follicular helper cells—shifting the peritoneal/TME cytokine balance away from IL-6/STAT3 and toward more effective antigen handling.[69] SCFA output from these taxa further down-modulates chemokines (CCL20) and IL-8 in tumors.	Promising
Microbiota supplementation via FMT (with Akkermansia-enriched donors or strain add-back)	Replaces a tumor-promoting consortium with a community that favors antitumor immunity. In ovarian-cancer models, supplementing *A. muciniphila* via FMT activates $CD8^+$ T cells and induces antitumor effects—likely via metabolic rewiring (glucose/AA flux) and dampening NF-κB/STAT3-linked cytokines.[70] Clinically, careful donor screening and standardized manufacturing are essential.	Under investigation
Intratumoral microbiota therapy (native or engineered bacteria delivered to, or homing in on, tumors)	Beneficial bacteria migrate/proliferate within tumors, rewiring glutamine/purine/pyrimidine metabolism; provide TLR agonism and metabolite delivery to ignite local immunity.[71]	Experimental
Increase SCFAs (dietary fiber/prebiotic strategies; targeted postbiotics)	Elevates butyrate/propionate/acetate at the gut-peritoneal axis. SCFAs act on epithelial and immune cells to suppress pro-inflammatory chemokines (e.g., CCL20) and IL-8 within tumors, helping to quell NF-κB/STAT3 signaling.[72] SCFAs also support effector T-cell metabolism and survival, aiding $CD8^+$ function noted in preclinical ovarian systems.[73] Pairing with probiotics/synbiotics may synergize.	Promising
Repurposed Drugs		
Metformin	Inhibits mitochondrial complex I (NADH dehydrogenase) activity and cellular respiration of electrons transported by glucose deprivation, stopping Krebs cycle function.[74][75] Multiple observational cohorts & meta-analyses suggest improved survival in OC users; prospective trials ongoing.	Promising
Itraconazole (antifungal)	Hedgehog/angiogenesis inhibition; impacts cholesterol trafficking & mTOR; may enhance autophagy blockade with HCQ.[76][77]	Under investigation

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